Evaluate Qwen2.5-VL on IAM Handwriting Dataset

Benchmarks CER and WER on Teklia/IAM-line (all splits).

> Runtime → Change runtime type → **T4 GPU**

In [5]:
```
!pip install -q jiwer datasets transformers torch torchvision qwen-vl-utils
```

```
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0:00:01
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00:00:0100:01
```

In [2]:
```
# Mount Drive and cache HuggingFace weights so they don't re-download each s
from google.colab import drive
import os
drive.mount('/content/drive')
os.environ["HF_HOME"] = "/content/drive/MyDrive/hf_cache"
print("HF cache:", os.environ["HF_HOME"])
```

```
Mounted at /content/drive
HF cache: /content/drive/MyDrive/cogs181/hf_cache
```

In [3]:
```
%%writefile evaluate.py
"""
Evaluate Qwen2.5-VL on IAM Handwriting Dataset

Loads the Teklia/IAM-line dataset from HuggingFace, runs each image through
the Qwen2.5-VL pipeline, and computes CER and WER against ground truth.
"""

import argparse
import csv
import time
from pathlib import Path

import torch
from jiwer import cer, wer


def main() -> None:
    parser = argparse.ArgumentParser(description="Evaluate Qwen2.5-VL on IAM
    parser.add_argument("--model_name", default="Qwen/Qwen2.5-VL-3B-Instruct
    parser.add_argument("--device", choices=["auto", "cpu", "cuda", "mps"],
    parser.add_argument("--num_samples", type=int, default=100, help="Number
    parser.add_argument("--output_dir", default="./Evaluation")
    parser.add_argument("--max_new_tokens", type=int, default=128)
    parser.add_argument(
        "--prompt",
        default="Transcribe the handwritten text in this image exactly. Outp
```

```
    )
    args = parser.parse_args()

    output_dir = Path(args.output_dir)
    output_dir.mkdir(parents=True, exist_ok=True)

    device = args.device
    if device == "auto":
        if torch.cuda.is_available():
            device = "cuda"
        elif torch.backends.mps.is_available():
            device = "mps"
        else:
            device = "cpu"
    print(f"Using device: {device}")

    from transformers import Qwen2_5_VLForConditionalGeneration, AutoProcess

    print(f"Loading model: {args.model_name}")
    t_load = time.time()
    processor = AutoProcessor.from_pretrained(args.model_name)

    if device == "cuda":
        model = Qwen2_5_VLForConditionalGeneration.from_pretrained(
            args.model_name,
            torch_dtype=torch.float16,
            device_map="auto",
            ignore_mismatched_sizes=True,
        )
    else:
        model = Qwen2_5_VLForConditionalGeneration.from_pretrained(
            args.model_name,
            torch_dtype=torch.float32,
            ignore_mismatched_sizes=True,
        )
        model = model.to(device)
    model.eval()
    print(f"Model loaded in {time.time() - t_load:.1f}s")

    from datasets import load_dataset, concatenate_datasets

    print("Loading Teklia/IAM-line dataset (all splits) ...")
    ds_dict = load_dataset("Teklia/IAM-line")
    dataset = concatenate_datasets(list(ds_dict.values()))
    print(f"Total samples in dataset: {len(dataset)}")

    if args.num_samples > 0:
        dataset = dataset.select(range(min(args.num_samples, len(dataset))))
    print(f"Evaluating on {len(dataset)} samples")

    predictions = []
    references = []
    csv_path = output_dir / "eval_results.csv"

    t_start = time.time()
    with open(csv_path, "w", newline="", encoding="utf-8") as f:
```

```
        writer = csv.DictWriter(f, fieldnames=["index", "ground_truth", "pre
        writer.writeheader()

        for idx, sample in enumerate(dataset):
            image = sample["image"].convert("RGB")
            ground_truth = sample["text"].strip()

            if not ground_truth:
                continue

            messages = [{"role": "user", "content": [
                {"type": "image", "image": image},
                {"type": "text", "text": args.prompt},
            ]}]
            text = processor.apply_chat_template(messages, tokenize=False, a
            inputs = processor(text=[text], images=[image], padding=True, re
            inputs = inputs.to(device)

            with torch.no_grad():
                generated_ids = model.generate(**inputs, max_new_tokens=args

            trimmed = [out[len(inp):] for inp, out in zip(inputs.input_ids,
            prediction = processor.batch_decode(
                trimmed, skip_special_tokens=True, clean_up_tokenization_spa
            )[0].strip()

            sample_wer = wer(ground_truth, prediction) if ground_truth else
            sample_cer = cer(ground_truth, prediction) if ground_truth else

            predictions.append(prediction)
            references.append(ground_truth)

            writer.writerow({
                "index": idx,
                "ground_truth": ground_truth,
                "prediction": prediction,
                "sample_wer": f"{sample_wer:.4f}",
                "sample_cer": f"{sample_cer:.4f}",
            })

            if (idx + 1) % 10 == 0 or idx == 0:
                print(f"  [{idx+1}/{len(dataset)}] GT: {ground_truth[:50]}

    elapsed = time.time() - t_start

    total_wer = wer(references, predictions)
    total_cer = cer(references, predictions)

    print("\n" + "=" * 50)
    print("EVALUATION RESULTS")
    print("=" * 50)
    print(f"Model          : {args.model_name}")
    print(f"Samples        : {len(references)}")
    print(f"WER            : {total_wer:.4f} ({total_wer*100:.2f}%)")
    print(f"CER            : {total_cer:.4f} ({total_cer*100:.2f}%)")
    print(f"Time           : {elapsed:.1f}s ({elapsed/max(len(references),1)
```

```
    print(f"Results CSV    : {csv_path}")
    print("=" * 50)

    summary_path = output_dir / "eval_summary.txt"
    with open(summary_path, "w", encoding="utf-8") as f:
        f.write(f"Model: {args.model_name}\n")
        f.write(f"Samples: {len(references)}\n")
        f.write(f"WER: {total_wer:.4f} ({total_wer*100:.2f}%)\n")
        f.write(f"CER: {total_cer:.4f} ({total_cer*100:.2f}%)\n")
        f.write(f"Time: {elapsed:.1f}s\n")
    print(f"Summary saved  : {summary_path}")


if __name__ == "__main__":
    main()
```

```
Writing evaluate.py
```

In [6]:

```
# Run evaluation (change --num_samples as needed, 0 = all)
!HF_HOME=/content/drive/MyDrive/hf_cache python evaluate.py --device cuda --
```

```
Using device: cuda
Loading model: Qwen/Qwen2.5-VL-3B-Instruct
preprocessor_config.json: 100% 350/350 [00:00<00:00, 2.83MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set
a HF_TOKEN to enable higher rate limits and faster downloads.
chat_template.json: 1.05kB [00:00, 6.11MB/s]
The image processor of type `Qwen2VLImageProcessor` is now loaded as a fast
processor by default, even if the model checkpoint was saved with a slow pro
cessor. This is a breaking change and may produce slightly different output
s. To continue using the slow processor, instantiate this class with `use_fa
st=False`.
config.json: 1.37kB [00:00, 8.42MB/s]
tokenizer_config.json: 5.70kB [00:00, 3.33MB/s]
vocab.json: 2.78MB [00:00, 126MB/s]
merges.txt: 1.67MB [00:00, 135MB/s]
tokenizer.json: 7.03MB [00:00, 162MB/s]
model.safetensors.index.json: 65.4kB [00:00, 232MB/s]
Downloading (incomplete total...): 0.00B [00:00, ?B/s]
Downloading (incomplete total...): 100% 7.51G/7.51G [00:25<00:00, 133MB/s]
Fetching 2 files: 100% 2/2 [00:26<00:00, 13.09s/it]
Download complete: 100% 7.51G/7.51G [00:26<00:00, 286MB/s]
Loading weights: 100% 824/824 [00:02<00:00, 319.66it/s, Materializing param=
model.visual.patch_embed.proj.weight]
generation_config.json: 100% 216/216 [00:00<00:00, 1.46MB/s]
Model loaded in 39.7s
Loading Teklia/IAM-line dataset (all splits) ...
README.md: 2.14kB [00:00, 10.5MB/s]
data/train.parquet: 100% 167M/167M [00:04<00:00, 36.3MB/s]
data/validation.parquet: 100% 24.7M/24.7M [00:01<00:00, 24.6MB/s]
data/test.parquet: 100% 73.6M/73.6M [00:01<00:00, 45.8MB/s]
Generating train split: 100% 6482/6482 [00:00<00:00, 20331.01 examples/s]
Generating validation split: 100% 976/976 [00:00<00:00, 18073.31 examples/s]
Generating test split: 100% 2915/2915 [00:00<00:00, 22549.77 examples/s]
Total samples in dataset: 10373
Evaluating on 10373 samples
  [1/10373] GT: put down a resolution on the subject  |  PRED: put down a re
solution on the subject
  [10/10373] GT: meeting of Labour 0M Ps tommorow . Mr. Michael  |  PRED: Me
eting of Labour 01/7 PR tomorrow. Mr. Michael
  [20/10373] GT: that the House of Lords should be abolished  |  PRED: that
the House of Lords should be abolished
  [30/10373] GT: that Labour should not take any steps  |  PRED: that Labour
should not take any steps
  [40/10373] GT: that the House of Lords should be abolished and  |  PRED: t
hat the House of Lords should be abolished and
  [50/10373] GT: Labour has to have an adequate number of  |  PRED: Labour h
as to have an adequate number of
  [60/10373] GT: the Federal Premier .  |  PRED: The Federal Premises.
  [70/10373] GT: have an adequate number of members . THE  |  PRED: have an
adequate number of members. THE
  [80/10373] GT: ( 280,000 members ) and Mr. Harry Nkumbula's  |  PRED: ( 28
0, 000 members ) and Mr. Harry Nhobula's
  [90/10373] GT: getting an elected majority in Northern  |  PRED: getting a
n elected majority in Northern
Rhodesia,
  [100/10373] GT: the talks fall through . There are bound  |  PRED: the tal
```

```
As he swung her on to the floor Gay saw Gavin
  [10120/10373] GT: may consider him a man , I would say he was a  |  PRED:
may consider him a man, I would say he was a
  [10130/10373] GT: glad to be able to occupy herself .  |  PRED: glad to be
able to occupy herself.
  [10140/10373] GT: " So glad that you are having such a lovely time ,  |  P
RED: "So glad that you are having such a lovely time, d
  [10150/10373] GT: to the man she had spent so much time  |  PRED: to the m
an she had spent so much time
  [10160/10373] GT: Because she was second best . The others came back  |  P
RED: Because she was second best. The others came back
  [10170/10373] GT: home , or perhaps a student studying to  |  PRED: home,
or perhaps a student studying to
  [10180/10373] GT: She was engaged to Nigel , had been for two years   |  P
RED: She was engaged to Nigel, had been for two years.
  [10190/10373] GT: the situation would have been intolerable  |  PRED: the
situation would have been intolerable
  [10200/10373] GT: of variety and it 's a change . " Nigel was the ho  |  P
RED: OF VAPOR AND IT'S A QUANLCE. " NIGEL WAS THE HOSPI
  [10210/10373] GT: dancing with a blonde at eight-thirty that  |  PRED: dan
cing with a blonde at eight-thirty that
  [10220/10373] GT: to somebody , Phil . He 's always doing that . "  |  PRE
D: to somebody, Phil. He's always doing that."
  [10230/10373] GT: handsome at other times .  |  PRED: hand some at other t
imes
  [10240/10373] GT: " Don't be such a baby ! " They were going  |  PRED: "Do
n't be such a baby! " They were going
  [10250/10373] GT: kissing somebody else's girl , " he decided wicked  |  P
RED: kissing somebody else's girl, " he decided wickedl
  [10260/10373] GT: then regarded the ring which he retrieved  |  PRED: Then
regarded the ring which he retrieved
  [10270/10373] GT: Sentence Database P03-189  |  PRED: Sentence Database
P03-189
  [10280/10373] GT: ' Well ! At least I know the truth .  |  PRED: Well! At
least I know the truth.
  [10290/10373] GT: coronary . And in that second  |  PRED: Coronary. And in
that second
the past back improvi
  [10300/10373] GT: when you reach your parents'  |  PRED: when you reach yo
ur parents'
  [10310/10373] GT: fided in her , given some expla-  |  PRED: fidedel lei,
giuven come expla-
  [10320/10373] GT: She went back to Philip . But  |  PRED: she went back to
Philip. But
  [10330/10373] GT: little joy , or satisfaction , it had  |  PRED: little j
oy, or satisfaction, it had
  [10340/10373] GT: to see what comes out at the inquest .  |  PRED: to see
what comes out at theujjist
  [10350/10373] GT: he 'll confine himself to the nervous  |  PRED: He'll co
nfine himself to the nervous
  [10360/10373] GT: patronizing word , ' she replied . ' I despise  |  PRED:
patronizing word,' she replied. 'Idepide
  [10370/10373] GT: the fake friend of the family , like the  |  PRED: The l
ate friend of the family, Eddie Lee

==================================================
```

```
EVALUATION RESULTS
==================================================
Model          : Qwen/Qwen2.5-VL-3B-Instruct
Samples        : 10373
WER            : 0.2926 (29.26%)
CER            : 0.0597 (5.97%)
Time           : 3269.6s (0.32s/sample)
Results CSV    : Evaluation/eval_results.csv
==================================================
Summary saved  : Evaluation/eval_summary.txt
```

In [7]:

```python
# Preview per-sample results
import pandas as pd
pd.set_option('display.max_colwidth', 80)
df = pd.read_csv('Evaluation/eval_results.csv')
print(f"Mean WER: {df['sample_wer'].astype(float).mean():.4f}")
print(f"Mean CER: {df['sample_cer'].astype(float).mean():.4f}")
df.head(20)
```

```
Mean WER: 0.2930
Mean CER: 0.0649
```

Out[7]:

	index	ground_truth	prediction	sample_wer	sample_cer
0	0	put down a resolution on the subject	put down a resolution on the subject	0.0000	0.0000
1	1	and he is to be backed by Mr. Will	and he is to be backed by Mr. Will	0.0000	0.0000
2	2	nominating any more Labour life Peers	nominating any more Labour Life Peers	0.1667	0.0270
3	3	M Ps tomorrow. Mr. Michael Foot has	MPs tomorrow. Mr. Michael Foot has	0.2857	0.0286
4	4	Griffiths, M P for Manchester Exchange .	Guiffiths, MP for Manchester Exchange.	0.7143	0.0750
5	5	is to be made at a meeting of Labour	is to be made at a meeting of Labour	0.0000	0.0000
6	6	A MOVE to stop Mr. Gaitskell from	A move to stop Mr. Gatsby from	0.2857	0.2727
7	7	0M P for Manchester Exchange .	OM P for Manchester Exchange.	0.5000	0.0667
8	8	A MOVE to stop Mr. Gaitskell from nominating	A MOVE to stop Mr. Gairthell from nominating	0.1250	0.0682
9	9	meeting of Labour 0M Ps tommorow . Mr. Michael	Meeting of Labour 01/7 PR tomorrow. Mr. Michael	0.5556	0.1739
10	10	any more Labour life Peers is to be made at a	any more Labour life Peer is to be made at a	0.0909	0.0222
11	11	Foot has put down a resolution on the subject	Foot has just down a revolution on the subject	0.2222	0.0667
12	12	and he is to be backed by Mr. Will Griffiths ,	and he is to be backed by Mr. Will Griffith,	0.1818	0.0435
13	13	which would appear to " prop up " an out-dated	which would appear to "prop up" an out-dated	0.4000	0.0435
14	14	Though they may gather some Left-wing support , a	Though they may gather some Left-wing support, a	0.2222	0.0204
15	15	life peers into existence , they should not	life goes into existence, they should not	0.3750	0.0930
16	16	and that Labour should not take any steps	and that Labour should not take any steps	0.0000	0.0000

	index	ground_truth	prediction	sample_wer	sample_cer
17	17	now put forward nominees . He believes	now put forward nominees. He believes	0.2857	0.0263
18	18	large majority of Labour M Ps are likely to	Large majority of Labour MPs are likely to	0.3333	0.0465
19	19	that the House of Lords should be abolished	that the House of Lords should be abolished	0.0000	0.0000

In [8]:
```
# Print summary
with open('Evaluation/eval_summary.txt') as f:
    print(f.read())
```

```
Model: Qwen/Qwen2.5-VL-3B-Instruct
Samples: 10373
WER: 0.2926 (29.26%)
CER: 0.0597 (5.97%)
Time: 3269.6s
```

In [9]:
```
# Copy results to Google Drive
import shutil, os
from google.colab import drive
drive.mount('/content/drive')

DRIVE_DEST = '/content/drive/MyDrive/Evaluation'
if os.path.exists(DRIVE_DEST):
    shutil.rmtree(DRIVE_DEST)
shutil.copytree('Evaluation', DRIVE_DEST)
print(f'Results saved to Google Drive: {DRIVE_DEST}')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, cal
l drive.mount("/content/drive", force_remount=True).
Results saved to Google Drive: /content/drive/MyDrive/Evaluation
```